

Program Structure & Programming Process

ELEC1006: ENGINEERING COMPUTING

Program Structure

- The structure of a program can be generally divided into three parts:
 - 1) Acquire input data:
 - from keyboard.
 - from files on a disk drives.
 - 2) Process input data.
 - 3) Output the processed input data to be:
 - displayed on screen.
 - written to a file.

Programming Process

1. Clearly define what the program is to do.
2. Visualize the program running on the computer.
3. Use design tools such as a hierarchy chart, flowcharts, or pseudocode to create a model of the program.
4. Check the model for logical errors.
5. Type the code, save it, and compile it.
6. Correct any errors found during compilation. Repeat Steps 5 and 6 as many times as necessary.
7. Run the program with test data for input.
8. Correct any errors found while running the program. Repeat Steps 5 through 8 as many times as necessary.
9. Validate the results of the program.

More info on C++

- [1] Learn Cpp.com: Tutorials to help you master C++ and object-oriented programming.
<https://www.learncpp.com/>
- [2] cplusplus.com
<https://cplusplus.com>